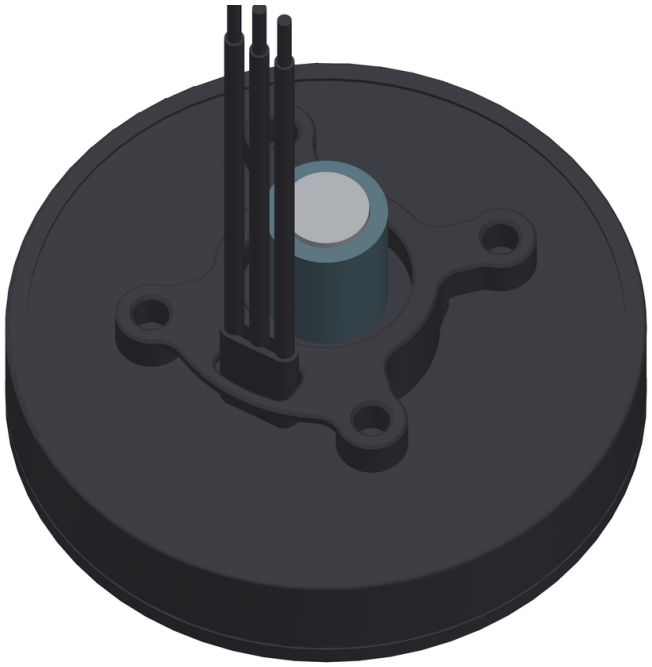
Magnets

How they work, how to fit them



Designed by Athanasia Nikolova

v1.1 · October 2026

All data come from tests and many real builds, not only from theory.

Teacher only

- **One thing per page**
  One action, one big picture. The part of the step is the only strong colour, the rest is quiet grey. Short sentences, much white space.
- **Colour is never the only clue**
  Every wire on a board drawing also has a number, every board a letter (X, Y, Z), every insert its size. The insert colours stay distinct with colour-vision deficiency (Okabe-Ito palette).
- **Readable**
  All instruction text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).
- **Print it right**
  In colour, at 100 % - not 'fit to page'. The Assembly guide draws every screw at real size: the student lays the screw on the page to check it.
- **Pace**
  One page, then the action, then the next page. For a student who tires quickly: one chapter per session. The check pages are good stopping points.

✋ Safety first

- Work with your teacher or mentor in the room.
- Strong magnets snap together: keep them apart, they pinch fingers.
- Super glue sticks skin in seconds. Use a toothpick, not your finger.
- Keep the magnets away from phones, bank cards and small children - a swallowed magnet is dangerous.

Two kinds of magnets

1. On each motor · ×3

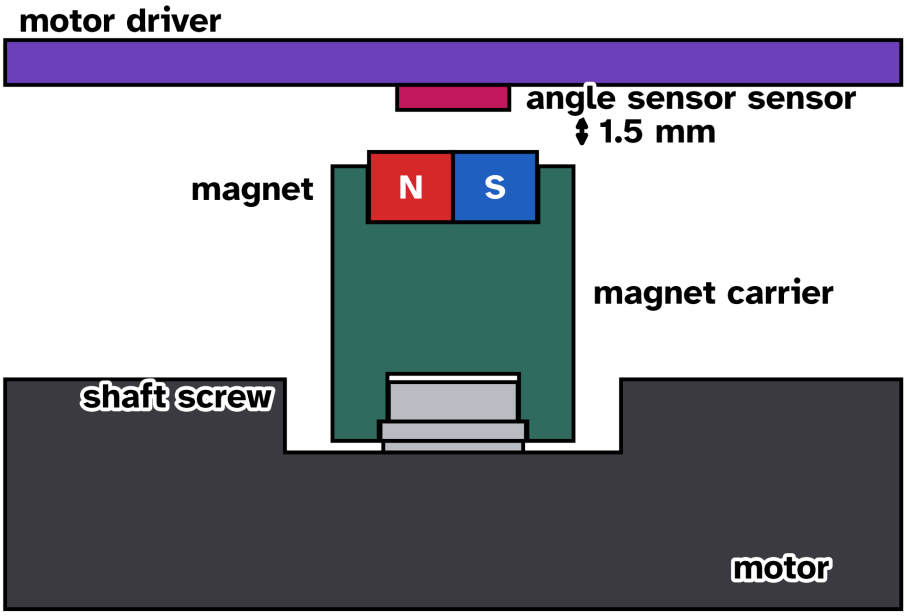
A round magnet Ø6 x 2.5 mm on the end of the motor shaft. It turns with the wheel. The motor driver reads it and knows the exact angle of the motor.

2. On the covers · ×48

Small magnets Ø3 x 2 mm. They hold the 3 covers on the box - no screws. Each cover magnet sits opposite a magnet in the box or in a beam.

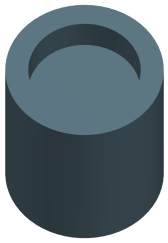
How the motor knows its angle

The magnet has a North half and a South half. The sensor under it feels where they point.



- The magnet must sit in the middle of the shaft - the handler keeps it there.
- The gap to the sensor is 1.5 mm. Nothing may touch.

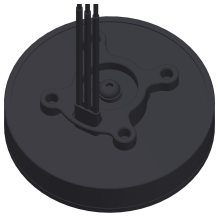
Motor magnets: you need



3 magnet carriers



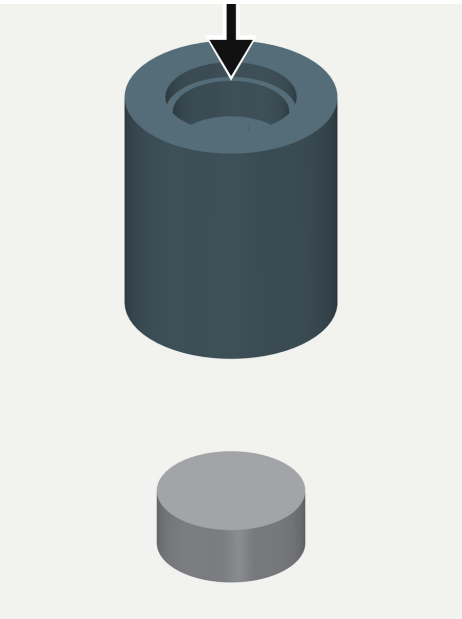
3 magnets Ø6 x 2.5



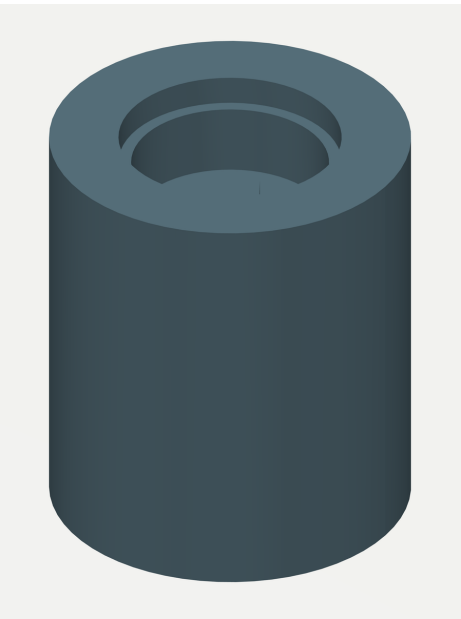
the 3 motors

- The magnet-handler (printed tool, magnet-handler.step)
- A hard, flat table
- Super glue (CA), a little - only for the carrier on the motor

1 Magnet into the carrier



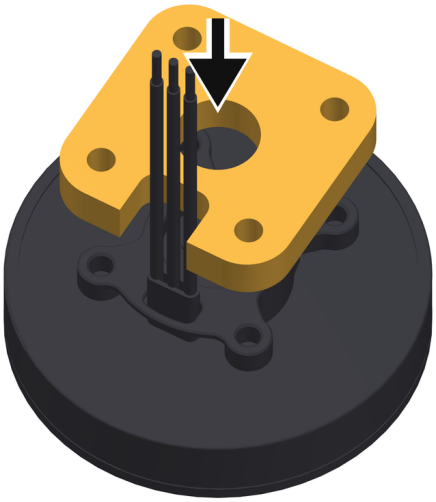
before



after

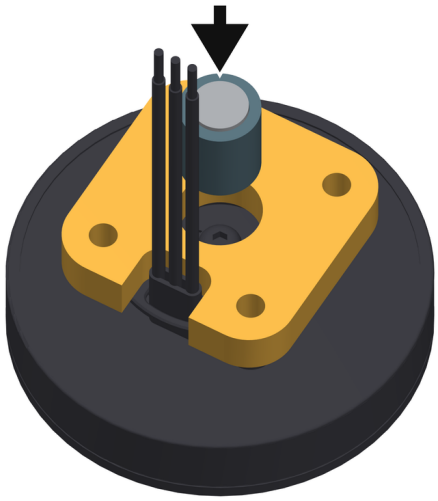
- Put the magnet flat on the table.
- Hold the carrier upside down over it, the round hole down.
- Press hard, straight down, until it stops. No glue.

2 Screw the handler on



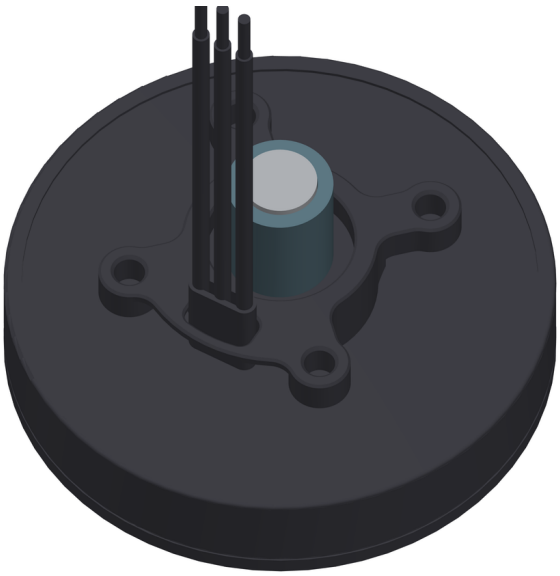
- Screw the magnet-handler onto the back of the motor.
- 4 M3 screws in the motor's own 4 holes. Its slot goes over the 3 motor leads.

3 Glue and press



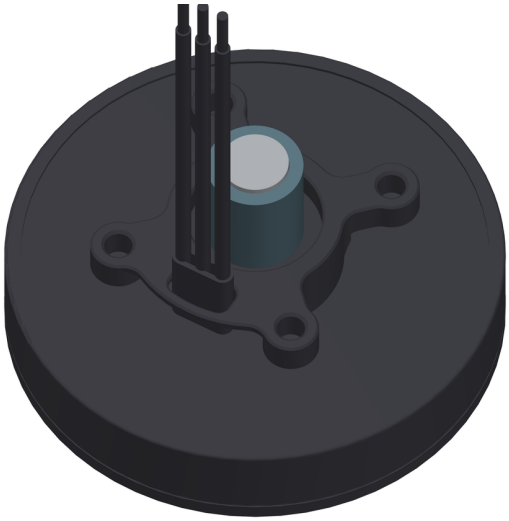
- A small drop of glue on the screw at the back of the motor.
- Push the carrier through the handler until it sits on the screw.
- Its little hex pin goes into the screw - do not turn it.

4 Wait, then take the handler off



- Leave it a few minutes: the handler keeps it straight while the glue sets.
- Unscrew the handler. The carrier stays, centred.

Check the motor magnet



- The carrier stands straight, the magnet is flat on top.
- Turn the wheel by hand: the magnet turns round, it does not wobble.
- Three motors - three magnets.

Cover magnets: where

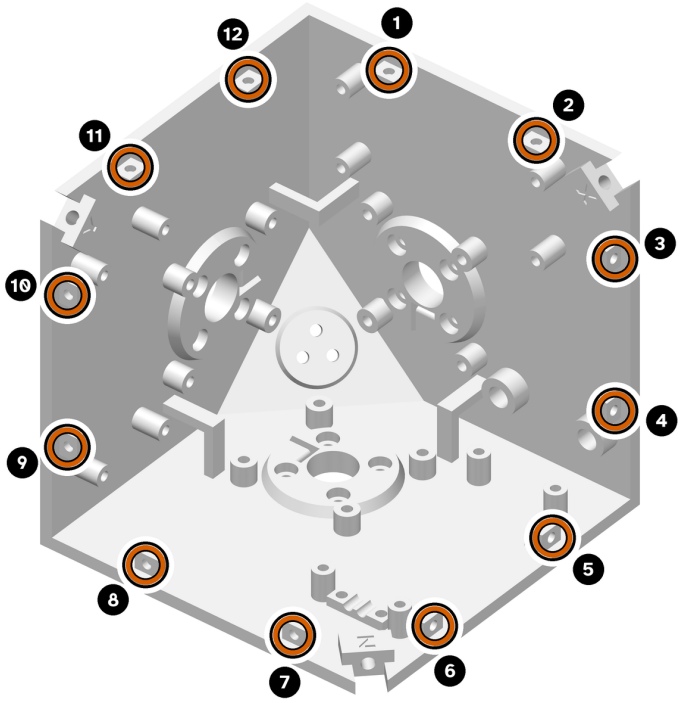
48 magnets, always in pairs.

In each cover	8 × 3 covers	**24**
In the box	12 (4 on each open face)	**12**
In each tie beam	4 × 3 beams	**12**
All together		**48**

- First all the box and beam magnets. Then the covers, one at a time.
- You need: super glue, a toothpick, a marker pen.

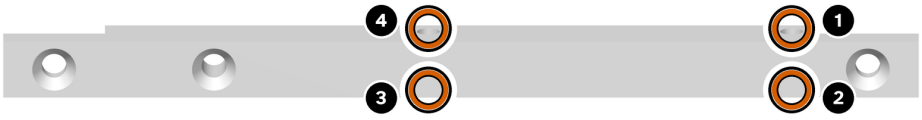
In the box · 12

4 small bases on the top of each open face.



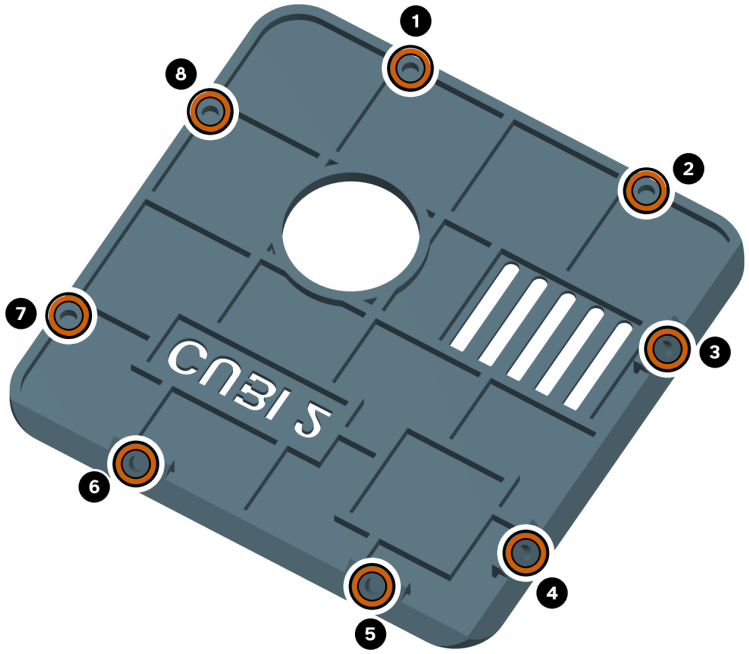
In each tie beam · 4

The 3 beams with the tie hole.

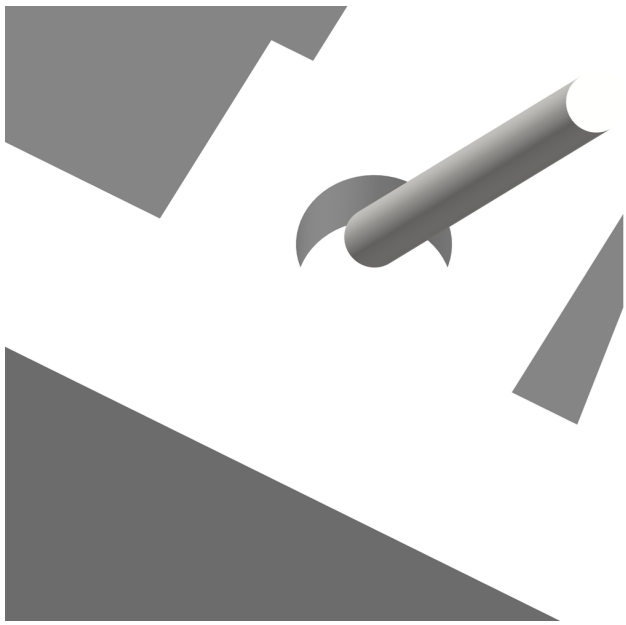


In each cover · 8

Inside face of the cover.

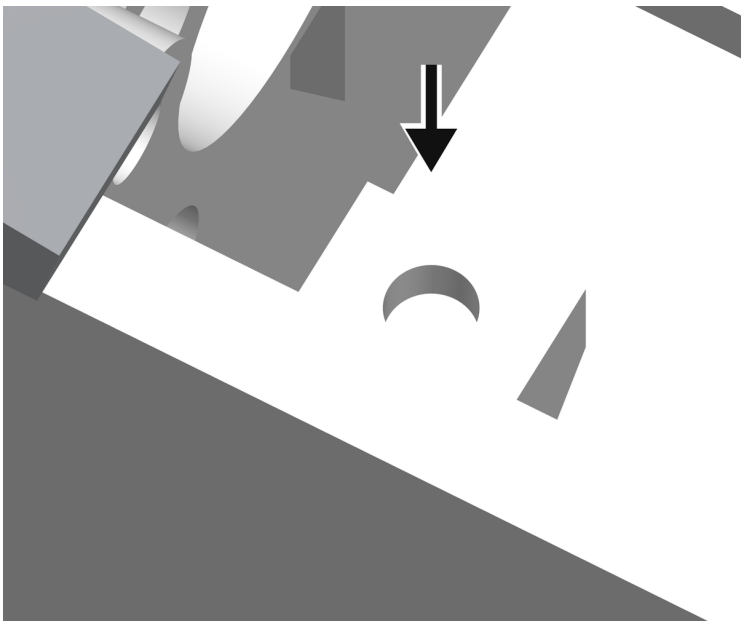


1 Glue in the pocket



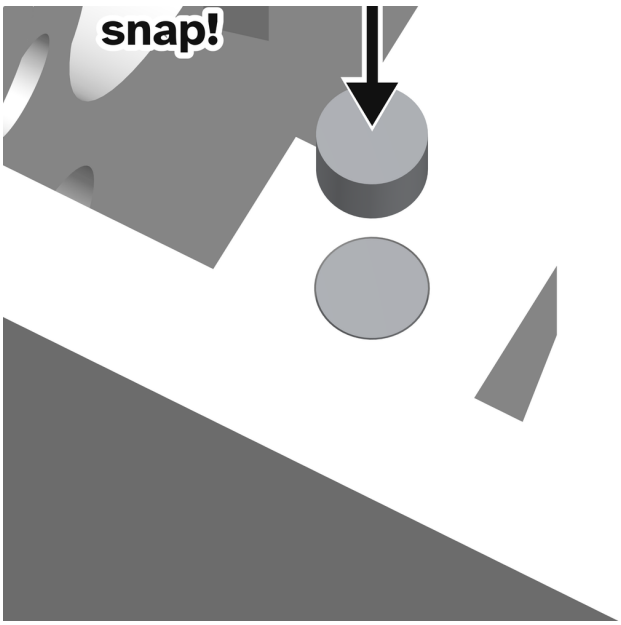
- A tiny drop of glue in the pocket, with a toothpick.
- Box and beams first - every pocket.

2 Magnet in, flat



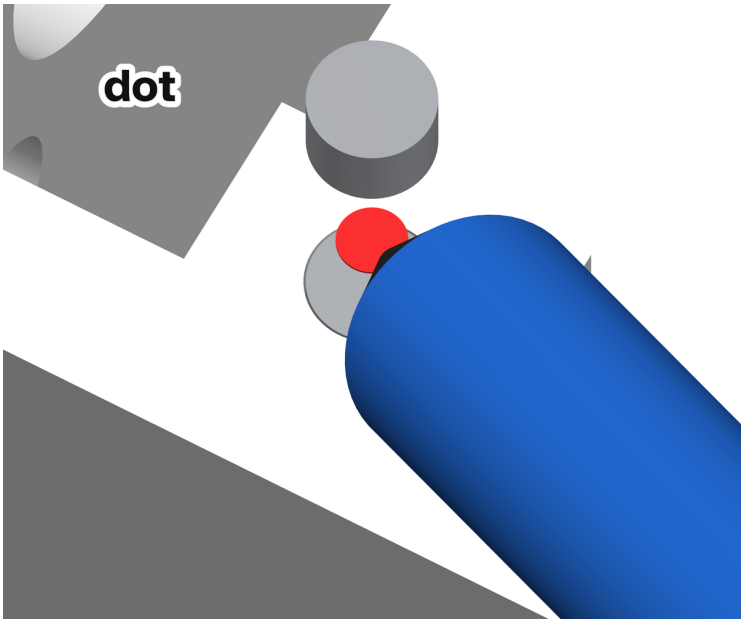
- Push the magnet in with a plastic or wooden stick - a steel tool pulls it out.
- Flat with the surface. Any way round here.

3 Let it snap



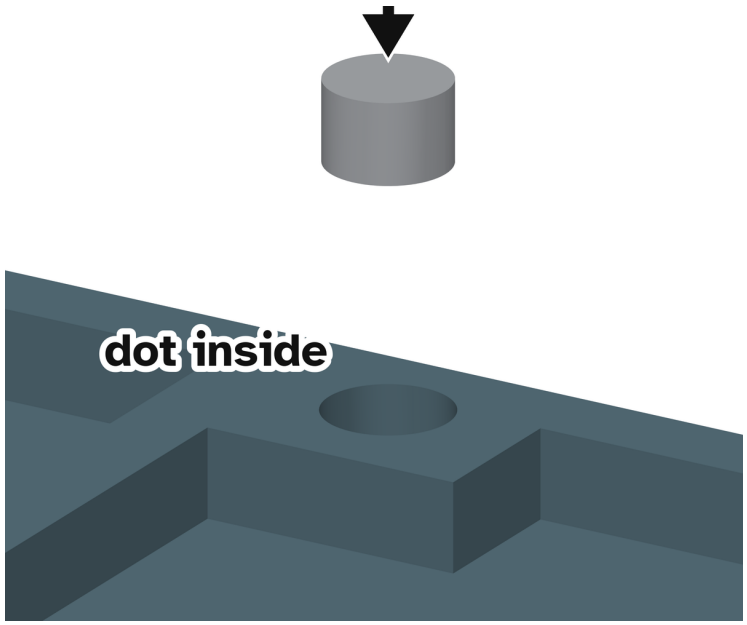
- Put a loose magnet on top of a glued one.
- It snaps on and turns itself the right way.

4 Mark its top



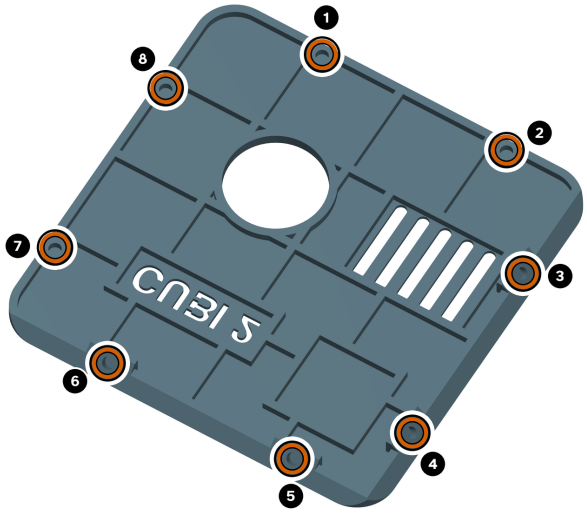
- A dot with the marker on its top face.
- Pull it off. The dot shows which face goes into the cover.

5 Into the cover - dot first



- A tiny drop of glue in the cover pocket.
- Push the magnet in with the DOT INSIDE, flat with the surface.
- Do the 8 pockets of this cover, each from its own partner.

6 Close the cover



- Wait for the glue, then put the cover on its face: it jumps into place.
- Write the face letter (X, Y or Z) inside the cover: it belongs there.

Good to know

- Glue only on the pocket - never on the face that touches another magnet.